

Monitoring the Future: Object and Source Memory for Real and Imagined Events

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SUMMARY

In these experiments a memory-monitoring decision is made, whereby subjects must decide not only whether or not to-be-learned stimuli will be remembered – the focus of all of the past research into the Judgement of Learning (JOL) – but also whether they will be able to assess the source of those stimuli, as assessed by a new measure, Judgement of Source (JOS). In Experiment 1 subjects had to judge whether they would remember the occurrence and the source of items that were either seen or imagined. Although seen items were better remembered and sourced than imagined, subjects were unable to predict this outcome: they underestimated their ability to recall seen items and overestimated their ability to recall imagined items. In Experiment 2 subjects had to discriminate between self-performed or other-performed enacted or imagined events. We expected that the motor cues associated with overt performance should provide more sensory information than had the visual input in Experiment 1, and this should help subjects to discriminate between real and imagined items. As predicted, JOL magnitude showed that subjects were now able to predict accurately that they would recall more enacted events than imagined events. JOS magnitude showed that subjects incorrectly predicted that self-enactment would assist source memory compared to imagination. However, it was the source of *other*-focused events which was more accurately remembered. The results are discussed in terms of Koriat's (1997) view about cue utility in making JOLs. Copyright © 1999 John Wiley & Sons, Ltd.

Research on eyewitness testimony and suggestibility has provided evidence for the important role of source monitoring, i.e. the ability to attribute the item we remember to the correct source. A common effect of suggestive misleading information is source misattribution, where people remember the item correctly, but then attribute it to the wrong source. We know that the information that people can use to remember that something occurred is not necessarily the same information they need to identify the source of the information (e.g. Johnson, 1988; Johnson and Raye, 1981; Lindsay and Johnson, 1991). We also know that when judging the future likelihood of remembering an item – a judgement of learning – a range of information is used (e.g. Nelson and Narens, 1990; Mazzoni and Nelson, 1995; Schwartz, 1994). What is

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not known, however, is whether the same information that is used to predict *that* something will be remembered is also used to predict *the source* of that event. That is, while we know that different types of information are used retrospectively to decide whether an item occurred, and to identify its source, it is not known whether different types of information are used prospectively to judge the likelihood of item and source remembering on a subsequent occasion. Therefore, the focus of the present experiments is on whether people can predict their ability to distinguish among various types of sources. Can people, for example, judge in advance whether they will be able to recognize when an item was the product of an internal source, and distinguish it from the case when it was the product of an external source?

This question can be nicely tested, and the result has potentially interesting implications. Consider the literature on suggestibility. There is increasing evidence that imagery is a powerful tool in changing people's memories. It was shown, for example that inviting people to imagine having spilled punch at a wedding produced detailed memories of the event in individuals who had previously stated that the event never happened to them (Hyman *et al.*, 1995). Thus imagination can produce a new and probably 'false' memory, the result of a source misattribution. People remember the content of the image (e.g. spilling the punch) correctly, but have lost the memory for the source of such content. Do people behave this way because they are not able to predict their ability to remember a mental image as a mental image? The studies reported here are a preliminary investigation of this topic. To our knowledge, there have been no previous studies of predictive judgements about one's ability to remember the origin of information, despite the fact that it is just as important to know the variables which people use to *predict* their memory for the origin of information as it is to know the factors that affect a person's ability to identify the original source of memorial information (Johnson *et al.*, 1993). Under some circumstances one has to judge whether one will in future be able to discriminate between sources of information, as, for example, when confidential information is imparted.

Numerous studies of metacognitive judgments have assessed people's ability to predict item memory (i.e. what they will be able to remember in a future memory test) but not source memory. Item memory predictions are called Judgements of Learning (JOLs), and are usually measured by asking people to rate the likelihood of their recall of various items, such as words or short sentences (Mazzoni and Cornoldi, 1993). When JOLs are then compared with actual recall, or recognition, JOL accuracy is above chance, and JOL magnitude is higher for recalled than for non-recalled items. In the present studies we want to see whether people are similarly accurate in predicting their ability to remember the source of such memory content. Consistent with the term 'Judgement of Learning (JOL)', these predictions hereinafter will be called JOSs, or Judgements of Source. As is the case for JOLs, there are two measures that can reveal whether subjects can discriminate in their predictions the source of the information. One is the magnitude of the prediction: we would expect to find a greater magnitude for the type of source that they will later be able to identify more correctly. The other is the relative accuracy of the JOS, indicating that when a prediction of source recognition is higher for one item compared to another, then source recognition for that item is also correct. If subjects are able to predict source recognition, we would expect to find that overall JOSs are as accurate as JOLs. In addition, an interesting aspect is to assess whether relative accuracy of JOS is higher for one type of encoding (e.g. imagery) than for another (e.g. visual).

A second aim of the study concerns the accuracy of JOL. Can people predict the different patterns of results usually obtained with different types of encoding, as in trying to encode a visual presentation versus creating a mental image of the item, versus enacting the motor activity to encode it? In Experiment 1 we manipulate imagining versus visual display of a picture. The issue of memory prediction of imagined versus visually presented items has been examined in some studies that have compared memory predictions when using an imagery strategy to encode, with memory predictions when using a rote repetition strategy (see Rabinowitz *et al.*, 1982; Shaughnessy, 1981). The results of these studies are mixed: in some cases predictions did not distinguish between imagined and rote-learned material, whereas in other cases subjects predicted higher recall for imagined than for repeated items. However, interactive imagery was used here; no previous studies have used a simple mental image of the item. Some variables (e.g. different kinds of encoding) can affect JOLs beyond an effect on the likelihood of recall. For instance, JOLs are more accurate after encoding via intentional learning than after encoding via incidental learning, even when the likelihood of recall does not differ for those kinds of encoding (Mazzoni and Nelson, 1995). While JOLs may differ among conditions, while recall does not, so also there are circumstances where JOLs are relatively insensitive to eventual marked differences in recall (Carroll *et al.*, 1997) which result from variables prevailing during learning.

Recently, Koriat (1997) has attempted to categorize cues which are used by subjects in making JOLs. One category, intrinsic cues, involve those which are inherent attributes of the material, such as its intrinsic imagery value, or its pre-experimental semantic characteristics. The second category, extrinsic cues, pertains to conditions of learning or encoding operations applied by the learner, such as number of repetitions of the material, level of processing, massed versus distributed learning, and interactive imagery. A third class of cues, mnemonic cues, refers to subjective phenomenal cues which are used as a source of information about the future memorability of material, such as the ease of processing the material or of the information coming to mind, the familiarity of the cue, or the pertinence of the material for oneself. These mnemonic cues are sensitive to both the intrinsic and extrinsic factors that influence judgements of learning.

Koriat's cue-utilization approach assumes that JOLs are inferential in nature. In general, Koriat predicts and finds support for the proposition that intrinsic factors have the *same* effect on JOL and recall, while extrinsic factors are *discounted* in predictions of recall relative to intrinsic factors. Both intrinsic and extrinsic factors can affect JOL directly, or indirectly through their influence on mnemonic cues. With practice, there is an increasing reliance on mnemonic internal cues in making JOLs, and a shift away from reliance on intrinsic factors.

In Experiment 1 our manipulation of imagined versus visually presented stimuli allows us to examine the effect of what is most likely an extrinsic factor on JOL and on JOS. The prediction is that this manipulation will affect recall (although it is not clear from the literature from which condition recall will benefit), but, following Koriat, that subjects will discount the condition of learning in making JOLs, and will not be able to predict that they will differentially remember items actually seen and items only imagined. An additional issue for examination is whether the same types of cues – intrinsic, extrinsic, or mnemonic – used for making JOLs also influence people's judgements about future source discrimination (JOSs). If so, a person may use

intrinsic factors, such as the *a priori* difficulty or memorability of the material, as a basis for judgements about source ('I'll remember that I saw that because it was such a complex picture'). It is also likely that people would use internal mnemonic cues to make such judgements ('That is not the sort of thing I would say, so it will be easy to determine that it was you who said it'). If JOSs utilize cues in the same way as do JOLs, it should be less likely that the extrinsic cue of type of encoding, as operationalized by seeing or imagining pictures, would be used in making prospective source judgements ('I'll remember that that was presented to me in pictorial form because I wouldn't have imagined that version of the object myself').

EXPERIMENT 1

This study requires subjects to both view visual stimuli (depictions of objects) and to imagine an object when given its verbal referent. We are interested in whether the *magnitude* of JOLs differs for 'seen' and 'imagined' stimuli. The study is also concerned with whether JOL *accuracy* is affected by the imagined and seen conditions; that is, whether items given high JOLs tend to be recalled, and items given low JOLs tend to be forgotten.

In addition, we investigate magnitude and accuracy of Judgements of Source (JOS) of the same items for which JOLs are made to determine whether subjects can monitor prospectively the items whose source they will later correctly remember, and discriminate these from those whose source they will confuse. According to Koriat, the experimenter-imposed conditions of learning (in our case, imagined versus seen) constitute an extrinsic influence on JOL which will be relatively discounted in predicting recall. Perhaps, too, JOS is influenced in the same way by extrinsic factors.

Method

Subjects

Forty undergraduate psychology students (mean age 20.6 years) were recruited; 34 females and 6 males. Participation was voluntary. The subjects were required to attend two sessions: a 20-minute acquisition session and a 15-minute test session one week later. Subjects were tested individually or in small groups of 4.

Materials and design

Subjects were exposed to a list which consisted of a random presentation of 25 pictures and 25 words for which they had to imagine mental pictures of objects. Both seen and imagined items represented common, everyday objects (e.g. 'bird', 'table'). The pictured version of each stimulus was a simple black on white line drawing. There were two lists of items: half the items were presented as pictures and half as words for half the subjects; for the other half of the subjects the format of the items was reversed. Twenty subjects were randomly allocated to Group 1 and received the Group 1 list of 50 items, and the remaining 20 subjects were allocated to Group 2 and received the Group 2 list of 50 items.

Procedure

An instruction sheet informed subjects that two kinds of stimuli would be presented to them: either a picture on an overhead projector or the name of an object spoken aloud. Pictures and words were intermixed, and there were five different random orders of item presentation, each of which was presented to one-fifth of the subject sample. For example, two pictures might be presented in succession, followed by the experimenter saying aloud a word which preceded a blank screen. The picture remained on the screen for 8 seconds for study. When a word was presented, subjects had 8 seconds in which to create a mental image of it, and were asked to mentally project their image onto the illuminated overhead projector screen. They were to imagine objects in the same form as the drawings, that is, as a simple line drawing. Immediately after they had either formed a mental image, or viewed a picture, they were to estimate the likelihood of freely recalling that item in one week's time. They recorded their judgement of learning on a rating scale which ranged from 0% to 100% (at 20% intervals), with 0% indicating that they would certainly not remember the item and 100% indicating that they certainly would remember the item. Their other task was to estimate how well they thought they would be able to discriminate the source of the item in a week's time. This judgement of source (JOS) was also recorded on the same type of rating scale, ranging from 0% to 100% with 0% indicating that they certainly would not know whether they had seen or imagined the item, and 100% indicating that they would certainly know whether they had seen or imagined it. A source judgement was not conditional on a recall judgement; if an item was judged as recallable with a 0% probability, subjects were asked to try to make an independent judgement with respect to source.

Subjects were aware that their memory for the items would be tested after one week, and that credit for their participation would only be given when they had completed the second session. After a period of one week, subjects returned to participate in the retention session. They were told to recall the names both of the pictures they were shown on the overhead projector, and the images which they had generated from the words. In both cases, the name of the item was to be written on the response sheet. Instructions urged them to try to recall as many items as possible, and to include an item even if they were relatively doubtful about it. Next to each recalled item subjects were to indicate whether the item had been presented or imagined, by writing 'seen' or 'imagined' beside it, in effect, forcing a source judgement for each item even if it were poorly recalled. They were allowed as much time as they liked (generally 15 minutes but no longer than 20 minutes). At the end of recall, subjects were also asked to briefly describe any learning strategies they had used during the acquisition phase.

Results

Recall was sourced as correct if the item produced at test had been presented during the learning session in either format. Mean proportions recalled for seen and imagined conditions are shown in Table 1. A paired *t*-test showed that a seen presentation format proved to be more effective than an imagined presentation format in assisting recall ($t(39) = 5.59$, $p < 0.01$). The false alarm rate was the proportion of occasions on which the subject recalled an item that had not been presented in either study format. These false alarms were also assigned to a 'source'. The mean

Table 1. Means and Standard deviations for all dependent measures in Experiment 1, for seen and imagined items

	Seen		Imagined	
	Mean	SD	Mean	SD
Magnitude JOL	50.6%	(14.1)	49.6%	(15.9)
Recall	0.60	(0.12)	0.40	(0.12)
Accuracy of JOL (gamma)	0.18	(0.41)	0.25	(0.46)
Magnitude of JOS	61%	(15.9)	58%	(16.7)
Correct source discrimination	0.76	(0.32)	0.63	(0.32)
Accuracy of JOS (gamma)	0.06	(0.44)	0.13	(0.49)
False Alarms	0.09	(0.92)	0.06	(0.78)

proportion of false alarms classified as seen and imagined, shown in Table 1, did not differ.

Source discrimination

Source monitoring was scored as correct if the item recalled was identified with the proper encoding condition. Of the 0.50 of the total items recalled, 0.70 were correctly sourced. Source discrimination was significantly above chance for both seen ($p < 0.01$) and imagined ($p < 0.05$) items (see Table 1). The mean proportion of items correctly sourced was greater for seen than for imagined items ($t(39) = 3.58, p < 0.01$).

Magnitude of JOL and JOS refers to mean judgement ratings only, whereas accuracy of JOL and JOS refers to the relationship between an individual item's judgement rating and eventual recall or non-recall of that item (for JOL), and eventual source attribution of that item (for JOS). The mean JOL ratings are shown in Table 1. Subjects underestimated the number of seen items they would recall by 10% and overestimated the number of imagined items they would recall by 10%. JOLs were no different for seen stimuli compared to imagined stimuli; subjects could not judge in advance whether they would remember more seen or more imagined items. In general, they believed that they would recall about half of the items regardless of whether they were seen or imagined.

Accuracy of JOL. JOL accuracy is operationalized as a Goodman–Kruskal gamma correlation between mean JOLs and mean proportion recalled for both seen and imagined stimuli (see Table 1). The gamma correlations were significantly different from zero (seen: $t(39) = 2.68, p < 0.01$; imagined: $t(38) = 3.41, p < 0.01$), indicating that subjects were above chance in their accuracy of prediction of recall performance, regardless of how the stimuli were presented. There was, however, no difference in accuracy of prediction for either type of stimulus.

Magnitude of JOS. The mean JOS ratings (see Table 1) did not differ for seen and imagined stimuli; subjects believed that they would discriminate the source of over half of the items irrespective of the encoding format. For both seen and imagined information, subjects underestimated what they would be able to correctly source, but in the case of seen information this underestimate was very great.

Accuracy of source discrimination. JOS accuracy was determined in the same way as JOL accuracy: by computing a gamma correlation between item-by-item JOSs and actual source discrimination for each subject. In neither case did the mean gamma scores shown in Table 1 differ from zero, showing that subjects were unable to predict their source discrimination ability for individual items at a rate above chance. There was also no difference between seen and imagined items with respect to source discrimination.

To further investigate the relationship between JOL and JOS, correlations were computed between individual subjects' ratings of magnitude of JOL and JOS for each type of stimulus format. This revealed a significant correlation ($r = 0.49$, $p < 0.01$) between the two measures for seen items, and a significant correlation ($r = 0.59$, $p < 0.01$) for imagined items. Subjects made similar judgements for memory and source discrimination regardless of presentation format. That is, a high JOL rating was associated with a high JOS rating. Perhaps, then, subjects used JOL to make their JOS ('If I can recall it, then I will recall its source').

We investigated the possibility that good recallers are also good source discriminators. There was no significant correlation between the proportion of seen or imagined information correctly recalled and the proportion of seen or imagined information correctly sourced, indicating that people who recall well have no greater likelihood of making a correct source discrimination than do people who recall poorly.

Post-experimental questioning revealed that subjects reported having used a number of different strategies to aid them in their learning of seen and imagined stimuli. The more common strategies included: making a mental picture of the seen stimuli, making an association between a seen picture and an item which they owned, categorising stimulus items into groups (e.g. animals, household items, body parts), silent repetition of items during learning, no explicit techniques, attaching a preference to some items, incorporating both seen and imagined items into a story scenario, and sequential learning.

In order to discriminate the source of stimuli, subjects reported: not trying to remember if they had seen or imagined them and instead focusing on item recall only; remembering a scene for imagined stimuli and imagining the item only for seen stimuli; visualization of pictures and no visualization for words (contrary to experimenter's instructions); using black and white imagery for pictures and colour for words. The inappropriateness of some of these strategies for both item recall and source recall may indeed be responsible for the lack of discrimination between seen and imagined stimuli in the magnitude of JOL and JOS responses.

Discussion

When subjects try to predict whether they will remember actually presented and imagined stimuli, and whether they will remember the origin of the items, their judgements do not differentiate between seen and imagined items. However, a week later, they are more likely to recall items which they saw than items which they imagined. Subjects underestimated their ability to recall seen items, and overestimate their ability to recall imagined items. Here, then, is a further example of cues which do not affect JOL but which do influence later performance (Mazzoni and Nelson, 1995; Carroll *et al.*, 1997). We find support for Koriati's hypothesis that extrinsic cues are discounted in making JOLs.

Furthermore, it is also the first study in which JOSs have been tested prospectively; these too appear not to be diagnostic of later source discrimination. Of those items which subjects recall, they are more accurate about the origin of the seen items than of imagined items, but they do not monitor this information when making their judgements. It is unclear whether failure of prospective source monitoring is due to the same types of cues being used and discounted for JOS as for JOL. While JOL and JOS magnitude showed the same pattern of discounting, (i.e. neither measure showed that magnitude was diagnostic of later remembering of either item or source) when item-by-item resolution was examined, accuracy was different for JOL and JOS: JOL accuracy was equal for both seen and imagined items, and there was above chance accuracy for both types of items. On the other hand, there was no significant accuracy for source monitoring.

Considering the calibration issue, why are people poor at monitoring and detecting that materials will be remembered better, following study, when they are actually presented than when they are imagined? One explanation is that proposed by Koriat: that extrinsic cues relating to the learning episode are discounted while intrinsic cues are overweighted. JOL resolution, as measured by the accuracy scores, was above chance, indicating that subjects were able to successfully predict that some items would be recalled and others would not. That is, there was some ability to discriminate between-item recallability. However, there was no overall advantage in resolution of one condition over the other, suggesting that whatever strategies were used in making such judgements, they did not capture any of the perceptual versus imaginal differences among the stimuli. This outcome is in accord with the conclusion by Koriat, that immediate JOLs are insensitive to extrinsic factors such as experimenter-imposed conditions of learning. Another explanation is that JOLs were made immediately after seeing or imagining each item, a situation in which JOLs are more inaccurate than when JOLs are delayed (Dunlosky and Nelson, 1994). The absence of the recommended delay in making JOLs in the present study was necessitated by the learning task: a single item was presented, and the judgement needed to be made then and there. There was no cue for each item which could be used as a marker for making each judgement some time after the initial learning; rather, the item itself would have to be shown a second time. Any judgement following this second presentation could, of course, have been based on the conditions surrounding the second presentation rather than the first. A third possibility is that the manipulation of seen versus imagined stimuli did not provide sufficiently distinctive cues to allow subjects to distinguish between modes of presentation, a point to which we return below.

The new finding of the experiment was that, not only are people poor at predicting whether they will better remember seen or imagined information, but they also are unable to predict with any accuracy the source of information which they will remember. The overall magnitude of JOS ratings suggests that JOSs do not tap any reliable information about source recognition, given that the actual source accuracy of the seen events was significantly greater than of the imagined events. The question is, on what are JOSs based, on JOLs? This failure of calibration is similar to JOL, and suggests that JOS may utilize the same cues as does JOL. Supporting this view is the significant correlation between people's JOL magnitude and JOS magnitude. Possibly, in the absence of good objective information for making source judgements, biases can influence memory monitoring (Mazzoni *et al.*, 1997).

Consistent with an alternative possibility, noted above for JOL, is the hypothesis that people may be poor at monitoring their ability to discriminate between imagined and seen information because of the nature of the material used in Experiment 1. Imagined items can be similar to seen items in some important ways: details, perceptual information, clarity, vividness, etc. That is, perhaps the two tasks were too similar to provide good discriminatory external cues which would allow accurate monitoring of item memory and source memory. In fact, we found subjects were not accurate in discriminating between seen and imagined items. They said 'seen' to 37% of the imagined items and 'imagined' to 24% of the seen items.

To test this 'indiscriminability' hypothesis we devised a new experiment. Experiment 2 addresses this issue by examining monitoring of imagined and seen information when the sources of information for the tasks are very different; in this case, when one involves motor information, and one visual information. In addition, we examine the effect of an intrinsic factor, encoding in terms of the self, on JOL and JOS.

EXPERIMENT 2

In Experiment 1 the requirement for subjects to discriminate between two similar visual encoding conditions may have been too difficult for accurate memory or source predictions. In Experiment 2 we require subjects to overtly perform or to covertly (imagine themselves) perform a task, or to watch or imagine others doing so. In this way, we achieve better discriminability between the task components; although enactment is a feature of all conditions, the motor cues associated with its overt performance should provide more sensory information than for covert enactment, and in turn this should help subjects to discriminate between real and imagined terms. It is known that items are better remembered if they are enacted by the subject rather than just heard (Engelkamp and Krumnacker, 1980), and this benefit is called the enactment effect. Anderson (1984) also found an enactment effect when comparing enacted versus imagined events, and others have found a benefit of enacting over listening, reading, planning and observing (Engelkamp and Zimmer, 1984).

Therefore, the first variable manipulated in this study is acquisition of the stimulus via enactment or imagining. When subjects are asked to make judgements about whether items will be easier to recall when covertly or overtly enacted, they will, according to Koriat's formulation, tend to discount the recall advantage for overtly enacted items. This is because we are assuming that the experimenter-imposed variable of enactment is an extrinsic cue, which they do not weight as heavily as a predictor of recall as they would weight, say, the intrinsic meaning of each item. That is, although there should be an advantage in later *recall* for enacted stimuli—the enactment effect described above—subjects should discount this variable in making their JOLs. The magnitude of JOLs will differ less than the actual recall difference between the two conditions.

In the present study, a second variable—the self—is manipulated as a source of information at encoding. Half of the items are to be performed or imagined by the subject him/herself and the other half by another. A number of researchers have noted that the self is implicated in the encoding and retrieval of memories (Conway, 1990; 1992, 1994; Conway and Bekerian, 1987; Conway and Rubin, 1993; Keenan and Baillet, 1980; Rogers *et al.*, 1997; Ross, 1989; Symons and Johnson, 1997) and the

preponderance of recollective experiences associated with encoding tasks which involve the self lead to superior memory (Symons and Johnson, 1997). We expect that encoding items in terms of the self will also affect source judgements. Ease and degree of elaboration of encoding of material in terms of the self should encourage higher judgements of source for self than for other-focused items, because subjects expect these richer cues to be available at recall later. Of course, whether these retrieval cues actually are as salient at recall as they are when subjects make source judgements at encoding is an empirical question. Nevertheless, people may *believe* that events involving themselves will be remembered better than events involving others.

Our assumption is that, in Koriat's terms, the self is either an intrinsic cue or a mnemonic cue for making Judgements of Learning, because it provides a rich source of idiosyncratic information. If classified as an *intrinsic* cue, information about the self should be *overweighted* when JOLs are made. If classified as a mnemonic cue, JOLs should accurately reflect later recall. A consequence of this is that memory-monitoring decisions should reflect the relative advantage of encoding in terms of the self compared to encoding in terms of another. Since encoding information in terms of the self is likely to confer a special salience on the action, self-encoding can be seen as similar to a variable such as the *a priori* semantic relatedness of two words. But it is not clear whether the intrinsic variable would have its effect directly on JOL, as does semantic relatedness (Carroll *et al.*, 1997) or indirectly through mnemonic cues.

Of interest, then, is whether predictions about memory for content and predictions about memory for source are affected in the same way by the variables of self versus other focus, and enactment versus imagining. Our inclination in Experiment 1 was to suspect that JOS simply reflects whatever influences JOL; however, with the manipulation of both an extrinsic variable—enactment versus imaging—which arguably allows for better source discriminability (see Johnson *et al.*) and an intrinsic variable—self or other focus—we allow any different influences on JOL and JOS to appear. It is also of interest to investigate whether accuracy of judgement of source (JOS) is similar to JOL.

Method

Subjects

Thirty-two undergraduate psychology students (average age 19 years) participated for course credit. There were 16 males and 16 females.

Design and materials

A 2×2 repeated measures design was used to examine the factors of Enactment and Focus. Each factor had two levels: performed versus imagined for Enactment and self versus other for Focus. The dependent variables were JOL magnitude, JOS magnitude, proportion of items freely recalled, and correct discrimination between performed and imagined actions.

A balanced Latin square sequencing design was used in presenting the items. The four sets of items were rotated through conditions. For example, four subjects performed the 20 actions in set A as the experimenter read out the actions, imagined themselves performing the actions in set B, watched the experimenter perform the actions in set C, and imagined the experimenter performing the actions in set D. The sequence of conditions was rotated systematically across groups of subjects. The

80 actions were selected from a larger set of actions used in previous research by Engelkamp and Zimmer (1984), on the basis that each action was clearly distinguishable from all others (e.g. scratch your nose; stand on one leg; make a fist; press the doorbell; strike a match).¹

Procedure

Subjects attended two sessions: acquisition and test. They attended the acquisition session in groups ranging from two to four and returned for testing exactly one week after the initial session. During acquisition dividing partitions were used to ensure that subjects were unable to see each other, while each could simultaneously see the experimenter. They were asked not to communicate with each other during the course of the experiment. When subjects performed the action, they did so with closed eyes, so as to reduce visual feedback. The same experimenter was used in all conditions so as to maintain a consistency of presentation.

At the beginning of the acquisition session, subjects were provided with a typed list of 80 actions organized into four sets, A, B, C and D. Adjacent to each action was a location for a source for JOLs and JOSs. Subjects were told that they would be asked by the experimenter to respond to a nominated action in one of four ways: perform the action themselves; witness the experimenter perform the action; imagine themselves performing the action; imagine the experimenter performing the action. Actions were to be exaggerated as in a mime performance and after 8 seconds of overt or covert performance, the experimenter would signal subjects to record scores on the score sheet, one for JOL and one for JOS. Both ratings consisted of a 6-point scale consisting of the intervals 0%, 20%, 40%, 60%, 80%, 100%. For JOL a score of 0% corresponded to the judgement that 'I will certainly not recall this item in a week's time' and 100% that 'I certainly will recall this item in a week's time'. For JOS, the corresponding judgements were 'I certainly will/will not remember whether this action was performed or imagined in a week's time'.

At test, the subjects returned to the laboratory, and were asked to write down any actions—either imagined or performed by the experimenter or the self—which they remembered from the previous session. As in Experiment 1, they were encouraged to guess, if they were unsure about whether an action had been presented. Next to each recalled action they were to indicate the source of the action, that is, whether it had been performed (by either the subject or the experimenter) or simply imagined. If subjects were unsure about the source of an action, they were to guess. After 15 minutes each subject had exhausted the number of actions which could be recalled.

Results and discussion

In judging recall data as correct or incorrect we did not adhere strictly to the verbal descriptors of actions; rather, if the action recalled was the same as the original in meaning, it was acceptable. For example, 'raise your leg off the ground' was acceptable for 'stand on one leg'; but 'open the door' was not interchangeable with 'close the door'. All analyses were 2×2 repeated measures analyses of variance on the factors of Enactment (performed/imagined) and Focus (self/other).

¹The full set of stimuli is available on request from the first author.

Table 2. Mean magnitude of Judgements of Learning (JOL) and Judgements of Source (JOS); propositions of items correctly recalled and discriminated, and JOL correlations, for all conditions in Experiment 2

	Self-acted	Other acted	Self-imagined	Other imagined
JOL	61% (18)	52% (21)	46% (17)	42% (20)
Recall	0.27 (0.15)	0.22 (0.10)	0.09 (0.07)	0.09 (0.06)
JOS	66% (17)	58% (20)	52% (18)	51% (22)
Discrimination	0.83 (31)	0.95 (14)	0.50 (41)	0.64 (41)
JOL accuracy	0.36** (0.44)	0.20* (0.52)	0.29* (0.60)	0.55** (0.43)

*Significant at $p < 0.05$, **significant at $p < 0.001$.

Recall

The proportion of items correctly recalled is shown in Table 2. The mean proportion correct recall for enacted items was 0.24 and the proportion for imagined items was very low at only 0.09. The difference was significant ($F(1,31) = 75.16$, $MSe = 100$, $p < 0.001$) and confirmed that an enactment advantage is readily found. There was no significant interaction ($F = 2.39$) and, contrary to our and the subjects' predictions, no advantage of self over other in recall ($F = 2.38$), with the mean proportion of self events recalled being 0.18 and the mean proportion of other events recalled being 0.15. The low proportion of items recalled in this study compared to Experiment 1 recall after a similar time interval suggests that these items constitute a more difficult set than those in Experiment 1, and this floor effect may limit the generalizability of our findings.

Source discrimination

In all conditions except the self-imagined condition, subjects' ability to discriminate source was significantly higher than chance ($p < 0.001$). When actual discrimination performance was examined for those actions which were recalled (and so this is a conditional measure based on only 0.17 of the total number of items), there was a significant enactment effect ($F(1,31) = 29.65$, $MSe = 0.11$, $p < 0.001$), such that the source of performed actions was indeed discriminated better than the source of imagined actions (means = 0.89 versus 0.57). Thus subjects were correct in their overall prediction that performance would assist source memory compared to imagination. There was also a significant effect of Focus, with the source of other-related actions discriminated more accurately than self-related actions (means = 0.79 versus 0.67) ($F(1,31) = 6.75$, $MSe = 0.08$, $p < 0.01$), an outcome which was directly contrary to what subjects had predicted. Thus there was more confusion about whether an item was performed or imagined when it was concerned with the self.

Judgement of Learning

We investigated how the mean magnitude of the JOL would be affected by the variables of Focus (self versus other) and Enactment. As the means in Table 2 show, there was a significant effect of Enactment ($F(1,31) = 33.47$, $MSe = 139$, $p < 0.001$); acting out the instruction gave rise to significantly higher ratings (mean = 56%) of likelihood of remembering than did imagining acting out the instruction (mean = 44%). Thus there was no support for the prediction that an extrinsic factor like enactment versus imagining would be discounted in making JOLs; subjects

believed that an enactment effect would occur. This suggests that in Experiment 1, the absence of monitoring in the seen versus imagined condition was due to the similarity of processing in the two conditions. When the conditions involved a motor and a visual task subjects did not discount the effect of the cues.

The Focus variable was believed to be a mnemonic cue and hence a rich source of idiosyncratic information. There was indeed a significant effect of Self versus Other on JOL magnitude (mean Self = 54%; mean Other = 47%) ($F(1,31) = 11.07$, $MSe = 142$, $p < 0.005$). The interaction failed to reach significance ($p = 0.10$).

Judgement of Source

The Focus variable had a significant effect on JOS ($F(1,31) = 4.92$, $MSe = 147$, $p < 0.05$), with subjects predicting they would discriminate source (performed or imagined) better for self-focused items than other-focused items (means = 59% versus 55%). The enactment variable was also significant ($F(1,31) = 14.26$, $MSe = 235$, $p < 0.001$); subjects believed they would discriminate source better for enacted events (mean = 62%) than imagined events (51.5%). More importantly, the interaction was significant ($F(1,31) = 5.29$, $MSe = 91$, $p < 0.05$) and is shown in Figure 1. Dependent t -tests showed that subjects were more likely to believe that they would discriminate the source when they performed the action themselves compared to when someone else performed it ($t(31) = 3.90$, $p < 0.001$). However, they could not make this judgement when considering imagined actions. They also believed that they would recall the source better when they themselves had performed the action compared to when they had imagined themselves performing the action ($t(31) = 4.33$, $p < 0.001$) but they did not think they would be able to discriminate between

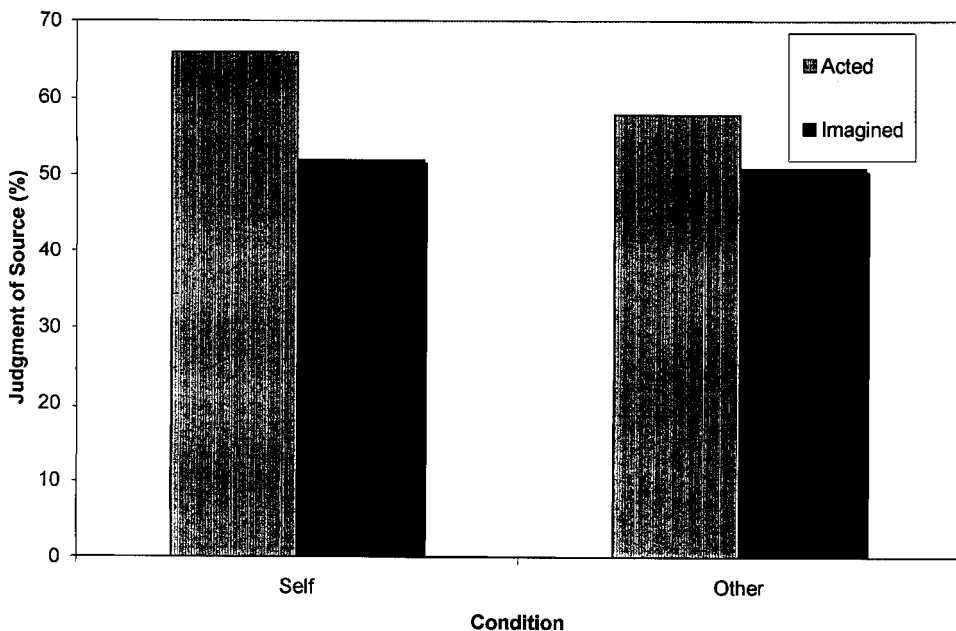


Figure 1. Judgements of source discrimination for actions either performed by the self or another or imagined being performed by the self or another, Experiment 2.

imagining another performing an action and watching another perform an action. In summary, the JOS magnitude shows a clear advantage for self-performed actions over any other type.

JOL accuracy

Accuracy of JOL was calculated using Goodman–Kruskal correlations for each subject, and tested for significance using a one-sample *t*-test. All coefficients in Table 2 were significantly above zero, showing that subjects could predict accurately which terms would be recalled. Analysis of the average gamma per subject was performed on only 20 of the 32 subjects, since gamma could not be obtained for at least one condition for 12 of the participants, due to empty cells resulting from the subjects failing to use all categories. When means from this subset were compared with those for the whole 32, a similar pattern emerged. In Figure 2, the mean scores are those of the subset on which the analysis was performed. There was a main effect of Enactment, ($F(1,19) = 5.63, p < 0.05$), and a significant interaction between Enactment and Focus ($F(1,19) = 5.68, p < 0.05$), as shown in Figure 2. *Post hoc* tests showed that when the focus of the actions concerned others, subjects more accurately predicted recall of items when they imagined rather than watched them performing the action ($t(19) = 3.81, p < 0.001$). On the other hand, when the items concerned the self, subjects were no more accurate at predicting recall of items which they themselves performed and imagined performing.

Accuracy of JOS was unable to be assessed since the gamma correlations were often unable to be provided and insufficient data were available. The low rate of recall for some conditions provided a low number of items to assess as being 'performed' or 'imagined'. Gamma was thus unable to be computed where so few data occurred.

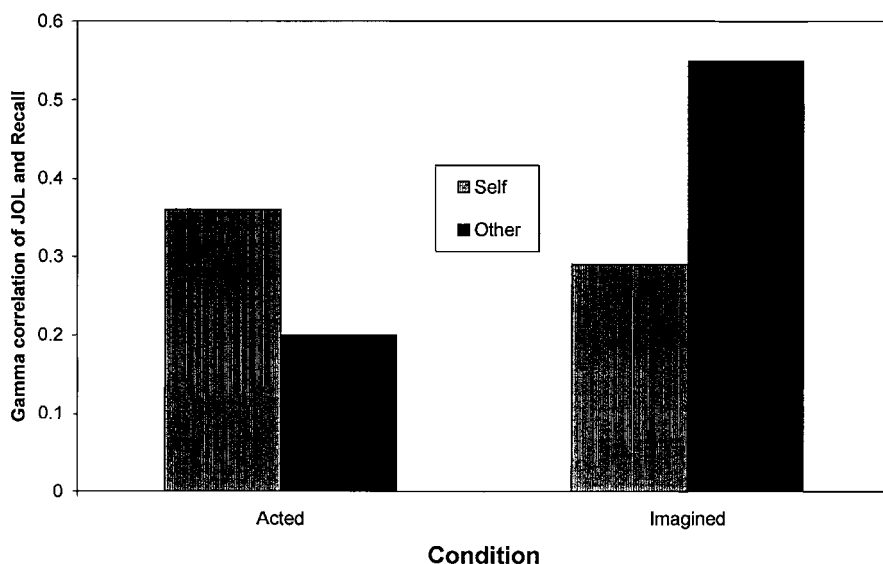


Figure 2. Accuracy of JOL: mean gamma scores obtained from a subset of the subjects in Experiment 2.

Discussion

In Experiment 2 subjects believed that actions which were actually performed would be recalled better than those which were only imagined. At recall a week later an enactment effect did indeed occur: there was superior recall when an action was performed compared to when it was only imagined. Experiment 2 also provides a clue to the reason for the absence of calibration of JOL in Experiment 1, where subjects did not predict that they would be able to recall more items which had been seen than items which had been imagined. It is likely that this 'relative discounting' of the influence of seen items on recall was due to the similarity of processing in the two conditions. When the task conditions became more discriminable, involving motor cues in the one and primarily visual cues in the other, the salience of the motor cues for JOLs was apparent.

With respect to the focus variable, encoding stimuli in terms of the self, an intrinsic variable, was predicted to lead the subject to use mnemonic cues of an idiosyncratic nature. Intrinsic cues are overweighted in making JOLs, argues Koriat, and we have found evidence for this in our study: Subjects believed that self-focused items would be better remembered than other-focused items, but this judgement was not validated by their later recall, where there was only a slight (but non-significant) advantage of self-focused over other-focused items.

Source judgements showed a different pattern from JOLs. For magnitude of JOS, but not for magnitude of JOL, there was an interaction between Enactment and Focus, indicating a clear belief by subjects that they would best remember source when the action was performed by themselves rather than another. On the other hand, they did not believe that source would be well remembered when actions were only imagined by themselves or another. There was an overweighting of the importance of self-performed actions, since the source of these actions was not, in fact, more accurately remembered than those in the other conditions. Although performance *per se* did in fact, help source memory, there was no difference between self-performed and other performed actions. What subjects certainly were unable to predict was that others' actions would be better sourced than actions focused on the self; in fact, they predicted the reverse. This is a potentially important finding; it means that, at least under the circumstances of this experiment, people are poorer at predicting discrimination between actions actually performed and actions only imagined when they themselves are the agents, whereas they are much better at predicting discrimination of source when someone else is the agent.

Comparing JOL and JOS magnitudes, the clearest difference is that in judging future item memory, subjects believe that performing, rather than imagining, actions will lead to better recall, and that self-focused items rather than other-focused items will also help recall, while in judging future source monitoring, they believe that it is self-enactment which will best help them to determine source. Thus JOS magnitude may be influenced in more subtle ways by combinations of variables. Overall, though, the magnitudes of JOL and JOS show similar patterns across conditions, but JOL greatly overestimates recall, consistent with other findings (e.g. Gigerenzer *et al.*, 1991; Griffin and Tversky, 1992) and JOS markedly underestimates source discrimination.

When object-level performance is prospectively monitored, predictions about both item recall and source recall memory one week later are not well calibrated. Item recall is much poorer than predicted, especially where the actions were imagined rather than

performed. This is consistent with the outcome of Experiment 1 where perceived items were recalled better than imagined items. Source recall is erroneously predicted to be better for self-focused items, whereas it is actually superior for other-focused actions, especially where the other performs the action. The superiority of the other-focused items is likely to be due to the peculiarities and variations of performance that render another's behaviour more memorable than one's own. Source recall is only at chance when one imagines oneself doing the action. It would seem that people overestimate the extent to which they will later be able to tell the difference between memories for actions they have performed and actions they have imagined. Although nothing is known about the basis for the JOS, it is quite likely that they are misled because the motor cues that seem so salient to them when performing the action are not available to them later when attempting to recall.

JOL accuracy showed an unexpected pattern of results. Although items were predicted with above chance accuracy in all conditions, it was the other imagined condition which resulted in the highest accuracy. A tentative explanation for this is that there were more 'degrees of freedom' when imagining another performing an action than in any other condition, and therefore items were more discriminable from each other. For instance, it is plausible that one can imagine another performing in a wider variety of ways than one can imagine oneself performing, simply as a result of experiences with the self. Similarly, when the items are enacted by either the self or another, there is much less room for variation between items. Although imagining, compared to perceiving, leads to poorer memory overall, then, imagining another's actions allows one to be more accurate in predicting which items will be recalled and which will not.

There are several implications for practice arising from these findings, implications which follow if, as has been shown (e.g. Nelson *et al.*, 1994), people can use their metacognitions to control their study and influence their learning; e.g. they can allocate their restudy effectively. First, if people erroneously believe that events involving themselves will be remembered better than events involving others, they may fail to pay as much attention to the details of self-focused events, with deleterious consequences. Just as monitoring success leads to controlled processing, so monitoring failure implies an absence of appropriate control. The overweighting of the importance of self for later recall is obviously not the result only of attending to feedback from motor cues at acquisition, since it was evident for the condition in which they merely imagined themselves performing the action. In this latter condition, the overestimate was particularly great, with recall only being 9% and JOL predicting 46% of the self-imagined items would be remembered. Possible reasons for why the self is thought to be such a rich vein of retrieval cues have been discussed by others (e.g. Conway and his colleagues) and termed the self-reference effect. Previous work on the role of the self in remembering suggests that ease of encoding events in terms of the self-schema leads to more elaborate, richer retrieval cues. These cues are likely to be particularly salient when the subject makes JOLs immediately following encoding. They are so salient that they lead to serious overestimation of how much will be remembered.

Second, when it comes to predicting judgements of source, the self plays an equally misleading role. Subjects are confident they will remember source between when the events involves self, particularly when the action is performed. However, it is the source of *other*-focused events which is more accurately remembered. Again, at retrieval some weeks later, the cues that were salient at acquisition, i.e. motor feedback

cues and ease of encoding, are no longer salient. What is more likely to determine source memory is variation and uniqueness of performance which is more likely to be associated with another than with oneself.

Finally, Koriat's formulation of the ways in which various cues influence JOL, although useful for categorizing some research findings, needs further refinement. There was no support for our prediction that enacting versus imagining an action is an extrinsic factor which is discounted in making JOLs. The conclusions to be drawn here are that either Koriat's formulation of the cues which affect JOL is incomplete or enactment is not an extrinsic variable. However, it does appear to fit Koriat's definition of such a variable, in that it is an experimenter-imposed condition of learning, where the encoding operations applied by the learner are varied. We recommend that there must be converging evidence that a cue is extrinsic or intrinsic independently of its effect on JOL.

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